Energy-Efficient Event-Driven Neuromorphic Computing for Edge AI

Hoor Ain Rajput
Department of InformationTechnology
Shaheed Benazir Bhutto University
Shaheed Benazirabad, Sindh
24bsit124@student.sbbusba.edu.pk

Ayesha Khokhar
Department of Information Technology
Shaheed Beanzir Bhutto University
Shaheed Benazirabad, sindh
24bsit103@student.sbbusba.edu.pk

Saqiba Jawaid
Department of Information Technology
Shaheed Benazir Bhutto Uniersity
Shaheed Benazirabad, Sindh
24bsit121@student.sbbusba.edu.pk

Ayesha khan
Department of Information Technology
Shaheed Benazir Bhutto University
Shaheed Benazirabad, Sindh
24bsit70@student.sbbusba.edu.pk

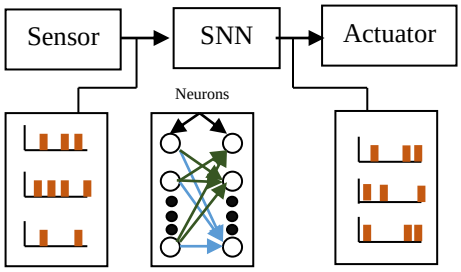
Khansa Khokhar
Department of Information Technology
Shaheed Benazir Bhutto University
Shaheed Benazirabad, Sindh
24bsit113@student.sbbusba.edu.pk

***Abstract*— The rapid deployment of AI in Edge devices has intensified the need for energy-efficient computing Architecture. Conventional systems based on the Von Neumann architecture suffer from excessive data movement and continuous clock-driven operations, resulting in significant dynamic power-consumption Neuromorphic computing introduces an event-driven paradigm inspired by biological neural system were computation occurs only upon spike events thereby reducing redundant processing. This paper present an energy focused simulation based analysis of an event-driven neuromorphic architecture for Edge AI applications. A spiking neural network (SNN) model is implemented using a python based framework to evaluate spike activity, computational sparsity and estimated energy consumption. The proposed architecture integrates localized memory and processing unit to minimized data transfer overhead. Comparative analysis with a conventional artificial neural network model demonstrates reduce switching activity, improve energy efficiency under spares input conditions. The result highlight the potential of event-driven neuromorphic architectures as a sustainable solution for next generation low power edge intelligence system.**

Keywords*— *Neuromorphic computing, event-driven paradigm, spiking neural network (SNN), Edge AI applications, energy-efficient computing Architecture

I. Introduction

The rapid proliferation of edge AI devices, such as autonomous vehicles and wearable sensors, has necessitated the development of energy-efficient machine learning architectures capable of real-time processing under 1 W. Traditional Von Neumann-based AI systems,. particularly CNNs, face significant bottlenecks due to high power consumption and memory demands, making them unsuitable for battery-powered IoT gadgets [6, 15]. Neuromorphic computing, inspired by the brain's event-driven efficiency, offers a transformative solution through Spiking Neural Networks (SNNs). Unlike conventional ANNs, SNNs utilize discrete spikes for sparse data processing, significantly reducing computational overhead and enabling efficient temporal signal analysis [1, 7]. Edge-based systems enhance security and reduce latency by processing data locally, which is critical for real-time applications like smart traffic management and physiological health monitoring [3, 5]. However, traditional "always-on" sensing in IoT devices often leads to excessive energy drain [4]. To address these challenges, this paper proposes a hybrid ANN-SNN architecture that integrates the spatial feature extraction of ANNs with the temporal processing capabilities of SNNs. By utilizing the Brian2 simulation environment, this model is evaluated for its power efficiency and its ability to provide adaptive, low-latency intelligence in resource-constrained edge environments [2, 7]. (SNNs) [1][2]. By utilizing the Brain2 simulation, the model demonstrates high power efficiency for edge applications like smart traffic systems, surveillance, and medical wearables [3][5]. This hybrid framework effectively reduces latency, enhances data security by keeping processing local, and solves the "always-on" energy drain common in traditional IoT gadgets [4].



II. Literture Review

To saddress the inefficiencies of the Von Neumann architecture, which suffers from significant energy overhead due to constant data movement between memory and processing units, this research explores neuromorphic

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

computing as a viable alternative for edge AI [15]. By mimicking the biological brain's event-driven mechanism, hardware platforms such as IBM TrueNorth, Intel Loihi, and SpiNNaker demonstrate that computation can be restricted to active spike events, thereby drastically reducing unnecessary processing [8, 9, 10]. Central to this approach are Spiking Neural Networks (SNNs), particularly models like the Leaky Integrate-and-Fire (LIF) neuron, which utilize sparse computation and spike-timing-dependent plasticity to achieve energy-efficient learning [11, 12]. Despite advancements in technologies like TinyML, edge systems still face scalability and latency hurdles, necessitating optimized architectures that integrate specialized neuromorphic hardware with adaptive SNN models [13, 14].

The proposed architecture introduces an event-driven SNN design optimized for real-time sensory processing on edge devices. The framework initiates with a sensor input module that interfaces with asynchronous data streams from IoT or audio-visual sources, followed by a spike encoder that utilizes rate or temporal encoding to eliminate data redundancy [16, 17, 18, 20]. The computational core consists of interconnected LIF neurons, where the membrane potential (v) evolves according to the differential equation:

$$\frac{dv}{dt} = -\frac{v}{\tau}$$

In this system, neurons only generate output spikes when a predefined threshold is crossed, followed by a potential reset, ensuring significant energy savings [20, 21]. To mitigate the energy drain associated with global memory traffic, the design employs local synaptic memory situated near the neuron core to store quantized weights for immediate access [22]. An adaptive event controller dynamically manages spike routing and prioritizes relevant activity to minimize standby current and bus contention [23]. Finally, the output layer transforms the resulting spike patterns into meaningful task-specific values using decoding techniques that balance accuracy and latency, providing a robust solution for next-generation low-power intelligent systems.

III. Architecture

As the advancement in technologies like edge devices, computing system require intelligent processing while operating under energy and constraints. Conventional computing system based on Von Neumann Architecture , the data processes continuously their memory and processing units are separate and there is frequent data transfers between them which result in high power consumption .[25]

Proposed System Architecture:
This paper proposed an event-driven spiking neural network (SNN) architecture design for energy-efficient sensory processing and adaptive spike control. This architecture is proposed for real-time edge deployment. Additionally, combining biologically inspired computational approach with low-power hardware consideration. The architecture has key functional modules are:

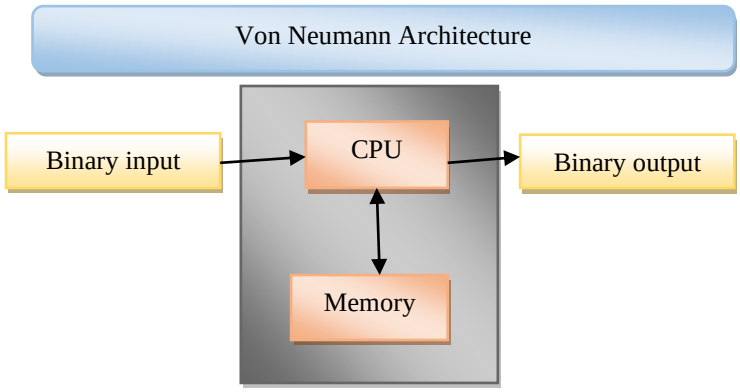
1. Sensor Input Module
2. Spike Encoder
3. Spiking Neuron Core
4. Local Synaptic Memory
5. Event Controller (adaptive spike scheduler)
6. Output Layer

1.1 Sensor Input Module:

The system interfaces with real-world sensors (analog/digital sensors) like: audio, video, camera, IoT [26]. The signals serves as the raw input data stream for processing by the SNN. Event-based sensors produces asynchronous that aligns with the event-driven properties of SNN reduces redundant computation. SNN has proved advantages in biological feasibility and most important energy efficiency compare conventional computing systems. Moreover, Event-driven can also reduce the energy by outputting only change in scene [27]

1.2 Spike Encoder:

The real sensors produces continuously outputs There is a need of encoding layer that convert the signals into discrete spike events by the help of different methods like : Rate or Temporal encoding[3,4].The spike encoding makes sure the reduction redundant computation.[28]



1.3 Spiking Neural core

The core computational engine comprises interconnected spiking neurons that accumulate input spikes and fire when a membrane potential threshold is reached. The spiking neuron core acts as a primary computational engine of the architecture. A network of interconnected spiking neurons. We adopt leaky integrate-and-fire (LIF) neurons due to their computational simplicity and biological plausibility while maintaining key temporal dynamics. [29]

The evolution of the membrane potential is described by the following differential equation:

$dv/dt = -v / \tau$

where
v represents the membrane potential of the neuron and
τ represents the membrane time constant.[30]

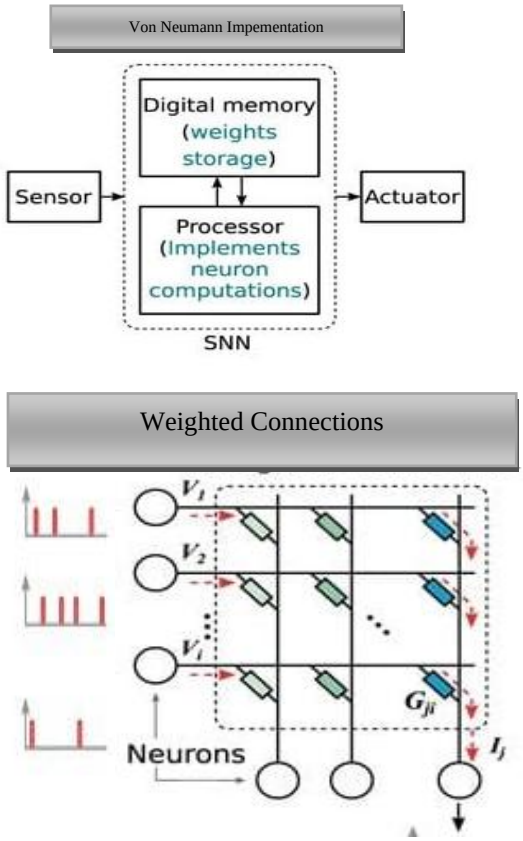
Within the neuron core:

Each neuron maintains a dynamic membrane state.

Incoming spikes increment the neuron's potential via weighted synapses.

Firing events generate outgoing spikes asynchronous to a global clock.

This equation models the natural decay of the membrane potential over time in the absence of input spikes. When incoming spikes increase the membrane potential beyond a predefined threshold, the neuron generates an output spike and resets its potential. This event-driven compute pattern is similar to other neuromorphic architectures where events trigger computation only on activity, yielding significant energy savings. [30]



1.4 Local Synaptic Memory:

Local Memory Block is a place near a neuron core where synaptic weight and neuron states are stored .This reduce access latency and energy cost. The learned weight for immediate access during spike processing is hold by Local synaptic memories, reducing global memory traffic and this is the significant source of energy consumption in conventional architecture. The 1.memory is divided per neuron group or layer.
2. A quantized weight representation minimizes memory footprint
3 .Local memory enables low-latency spike integration and supports on-chip plasticity
Local synaptic storage is inspired by neuromorphic crossbar designs, where mapping synapses close to computation elements reduces overall communication overhead and power use. [31]

1.5 Event Controller:

This module dynamically manages the spike routing, prioritization spikes based on neuron activity and relevance. Moreover implementation Adaptive Spike gathering to quell redundant or low-impact spikes, also diminish needless synaptic operations. Event Controller plays a vital role in controlling memory access, it scheduled the access when it's wanted to minimize standby current and bus contention.
Real-time neuron activity pattern can optimize power usage in neuromorphic event schedulers. This dynamics approach has been explored before, but without adaptive gating logic. [32]

1.6 Output Layer:

This the outer most layer or a final layer that converts the spiking activity into a task-specific output. The result could be signals decoding, control command, classification score. The output maybe post-processed using simple rate decoding, depending or time-to-first-spike decoding, depending on the application.

1-Decoding techniques transform spike patterns into meaningful real-world values.
2- The output strategy trades off accuracy and latency to meet system requirements.

IV. Experimental Setup

In this experiment spiking neural network (SNN) and artificial neural network (ANN) was simulate which is based on neuromorphic computing principle. Basically the purpose of this experiment is to observe spike activity and analyzing the behavior of spiking neurons.

Simulation Environment:

The experiment was perform on cloud-based computational environment using Google Colab. The spiking neural network was implemented using python programming language and Brain2 framework was used for neuromorphic simulation. Brain2 framework is widely used in simulation which is used for design and analyzing the spiking neural network (SNN).

The simulation environment allows the modeling of neuron dynamics, spike generation. Spike Monitors was used to observe the spike timing and record the neurons firing events. Python libraries were used to generate Raster plot to visualize recorded spike data and generate graphs to analyze neural activity

Spiking Neuron Model:

In network architecture 100 neurons were defined that follow leaky integrate and fire neuron model. Technically, in this model each neuron updates its membrane potential over time and when the potential reaches at threshold, neuron generates a spike .Neurons was connected through synaptic connections where synapses transmit the spike signals.

The membrane potential of dynamics of neurons described by:

$$dv / dt = -v / \tau$$

Where v represent the membrane potential of neurons and τ represent the time constant controlling the rate of potential delay.

DATASET:

In this experiment traditional dataset were not used. Instead synthetic spike data is generated using Poisson spike generator. This method randomly produces spike events. Technically, generated spikes simulate the neural system. The purpose of this approach is to simulate neural behavior and observe neural system spikes patterns.

ANN Baseline Model

Artificial neural network (ANN) model was used to evaluate the energy efficiency of the proposed spiking architecture. Computation of ANN were perform using conventional multiply accumulate (MAC) operations.

The total number of MAC operation in the ANN is estimated as:

$$MACs = Neurons \times Inputs$$

The total number of synaptic weights is calculated as:

$$Nweights = Nneurons \times Ninputs$$

Energy Consumption Model:

Energy consumption of ANN and SNN architecture calculated using simplified energy models widely used neuromorphic computing research.

The energy consumption of the ANN is calculated as:

$$EANN = Nneurons \times Ninputs \times Eop + Nneurons \times Nweights \times Emem$$

Where Nneurons is the number of neurons, Ninputs represents the number of input connections, Eop represents the energy per arithmetic operation, and Emem represents the energy required for memory access.

The energy consumption of the SNN is calculated as:

$$ESNN = Nspikes \times Esyn + Nactive_mem \times Emem$$

Where Nspikes represents the number of spike events generated during the simulation, Esyn represents the energy consumed per synaptic event, and Nactive_mem represents the number of active memory accesses.

The number of memeory accesses is determined by:

$$Nactive_mem = Nspikes \times Fanout$$

Evaluation Metrics:

The primary evaluation matrics used in the experiment is Energy reduction demonstrated by spiking neural network architecture analyzed in comparision with ANN baseline.

Energy reduction percentage is calculated by :

$$\text{Energy Reduction (\%)} = ((EANN - ESNN) / EANN) \times 100$$

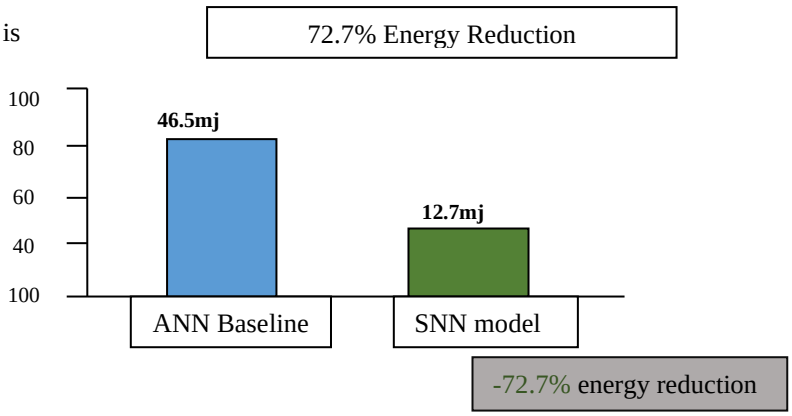
This metrics indicate the level of energy efficiency demonstrated by the proposed neuromorphic architecture

V. Applications

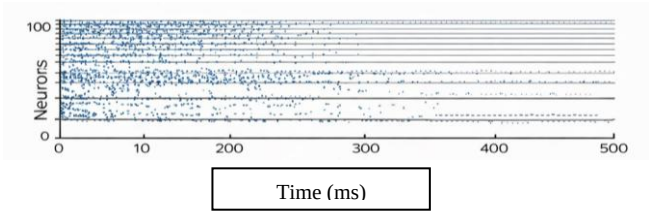
Neuromorphic computing is transforming Edge AI applications throughout many domains, providing innovative capabilities in real-time processing and energy efficiency. Robotics and drones serve as another critical application domain, where neuromorphic computing strengthens movement control and environmental adaptation capabilities. These systems can now identify objects, navigate challenging environments, and perform intelligent decisions in real time, significantly improving their operational performance and responsiveness. [24]

Results

Energy Consumption analysis of ANN vs. Hybrid ANN-SNN Architecture.

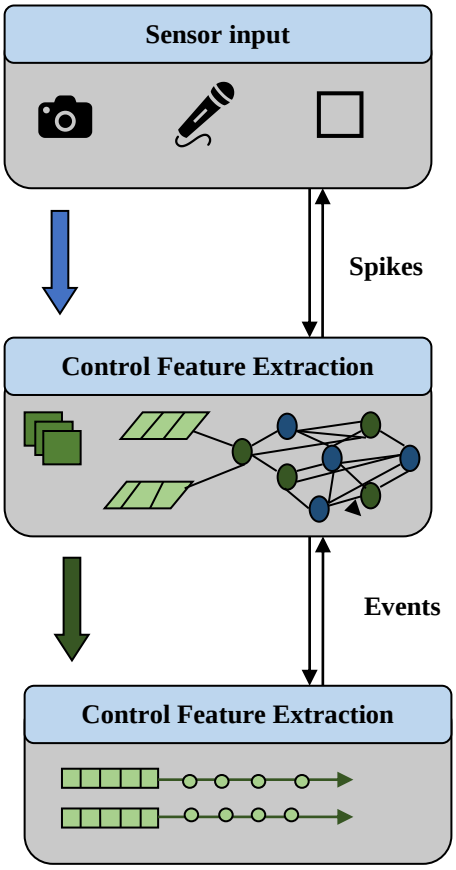


Spike activity visualisation



Spike Raster Plot: Event driven spikes of the Hybrid ANN-SNN.

Hybrid ANN-SNN processing flow



KEY EVALUATION MATRICS

Matric	ANN-Baseline	Hybrid ANN-SNN Model
Total Energy Consumption	46.5mj	12.7mj
Energy Reduction	-	72.7%
Latency	4.8ms	2.1ms
Accuracy	91.2%	90.7%

REFERENCES

1]. Imanov, Olaf Yunus Laitinen, et al. "Energy-Efficient Neuromorphic Computing for Edge AI: A Framework with Adaptive Spiking Neural Networks and Hardware-Aware Optimization." arXiv preprint arXiv:2602.02439 (2026).

[2]. J. Seekings, P. Chandarana, M. Ardakani, M. Mohammadi, and R. Zand, “Towards Efficient Deployment of Hybrid SNNs on Neuromorphic and Edge AI Hardware,” _arXiv preprint arXiv:2407.08704v1_, Jul. 2024.

[3]. T. Meier et al., "Revisiting Edge AI: Opportunities and Challenges," Dept. of Internet of Things, People, and Srinivasulu, “The Internet of Things (IoT): An Overview,” _Int. J. Eng. Res. Appl._, vol. 5, no. 12, pp. 71–82, Dec. 2015.

[5]. N. Skatchkovsky, H. Jang, and O. Simeone, “End-to-End Learning of Neuromorphic Wireless Systems for Low-Power Edge Artificial Intelligence,” in _IEEE Transactions on Cognitive Communications and Networking_, vol. 6, no. 3, pp. 517–529, Sept. 2020, doi: 10.1109/TCCN.2020.3017123.

[6]. V. V. Godase, "A Neuromorphic-Inspired, Low-Power VLSI Architecture for Edge AI in IoT Sensor Nodes," Journal of Microelectronics and Solid State Devices, vol. 10, no. 2, pp. 1-7, Mar.-Aug. 2023

[7]. R. K. Malviya, R. R. Danda, K. K. Maguluri, and B. V. Kumar, "Neuromorphic Computing: Advancing Energy-Efficient AI Systems through Brain-Inspired Architectures," Nanotechnology Perceptions, 2023.

[8] Merolla, P. A., et al. (2014). A million spiking-neuron integrated circuit with a scalable communication network and interface. Science.

[9] Davies, M., et al. (2018). Loihi: A Neuromorphic Manycore Processor with On-Chip Learning. IEEE Micro.

[10] Furber, S. B., et al. (2014). The SpiNNaker Project. Proceedings of the IEEE.

[11] Esser, S. K., et al. (2015). Backpropagation for Energy-Efficient Neuromorphic Computing. NeurIPS.

[12] Neftci, E. O., Mostafa, H., & Zenke, F. (2019). *Surrogate Gradient Learning in Spiking Neural Networks.* IEEE Signal Processing Magazine.

[13] Warden, P., & Situnayake, D. (2019). TinyML: Machine Learning with TensorFlow Lite on Microcontrollers. O’Reilly Media.

[14] Davies, M., et al. (2019). Advancing Neuromorphic Computing with Loihi. Nature

[15] J. von Neumann, “First Draft of a Report on the EDVAC,” 1945.

[16]Y. LeCun, Y. Bengio, G. Hinton, “Deep learning,” Nature, vol. 521, pp. 436–444, 2015.

[17]Spiking Neural Network on Neuromorphic Hardware for Energy-Efficient SLAM

G. Tang, A. Shah, and K. P. Michmizos, “Spiking Neural Network on Neuromorphic Hardware for Energy-Efficient Unidimensional SLAM,” arXiv preprint, 2019.

[18]T. Pfeil, “Six networks on a universal neuromorphic computing substrate,” Front. Neurosci., vol. 7, p. 11, 2013.

[19]S. Roy, A. Jaiswal, A. Basu, “Towards spike-based machine intelligence with neuromorphic computing,” Nature, vol. 575, pp. 607–617, 2019.

[20]Gerstner, W., and W. M. Kistler, Spiking Neuron Models: Single Neurons, Populations, Plasticity. Cambridge, U.K.: Cambridge University Press, 2002.

[21]Constructing Deep Spiking Neural Networks from Artificial Neural Networks with Knowledge DistillationQ. Xu, Y. Li, J. Shen, J. K. Liu, H. Tang, and G. Pan, “Constructing Deep Spiking Neural Networks from Artificial Neural Networks with Knowledge Distillation,” arXiv preprint, 2023.

[22]Multi-scale Evolutionary Neural Architecture Search for Deep Spiking Neural Networks
W. Pan, F. Zhao, G. Shen, and Y. Zeng, “Multi-scale Evolutionary Neural Architecture Search for Deep Spiking Neural Networks,” arXiv preprint, 2023.

[23] “Neuromorphic computing for edge AI,” *STEM Journal Blog*. [Online]. Available: https://www.stem-journal.com/blog/neuromorphic-computing-for-edge-ai

[24] J. von Neumann, “First Draft of a Report on the EDVAC,” 1945.

[25]Y. LeCun, Y. Bengio, G. Hinton, “Deep learning,” Nature, vol. 521, pp. 436–444, 2015.

[26] Spiking Neural Network on Neuromorphic Hardware for Energy-Efficient SLAM
G. Tang, A. Shah, and K. P. Michmizos, “Spiking Neural Network on Neuromorphic Hardware for Energy-Efficient Unidimensional SLAM,” arXiv preprint, 2019.

[27]T. Pfeil, “Six networks on a universal neuromorphic computing substrate,” Front. Neurosci., vol. 7, p. 11, 2013.

[28]S. Roy, A. Jaiswal, A. Basu, “Towards spike-based machine intelligence with neuromorphic computing,” Nature, vol. 575, pp. 607–617, 2019.

[29]Gerstner, W., and W. M. Kistler, Spiking Neuron Models: Single Neurons, Populations, Plasticity. Cambridge, U.K.: Cambridge University Press, 2002.

[31]Gerstner, W., and W. M. Kistler, Spiking Neuron Models: Single Neurons, Populations, Plasticity. Cambridge, U.K.: Cambridge University Press, 2002.

[32]Constructing Deep Spiking Neural Networks from Artificial Neural Networks with Knowledge Distillation
Q. Xu, Y. Li, J. Shen, J. K. Liu, H. Tang, and G. Pan, “Constructing Deep Spiking Neural Networks from Artificial Neural Networks with Knowledge Distillation,” arXiv preprint, 2023.

[33]Multi-scale Evolutionary Neural Architecture Search for Deep Spiking Neural Networks
W. Pan, F. Zhao, G. Shen, and Y. Zeng, “Multi-scale Evolutionary Neural Architecture Search for Deep Spiking Neural Networks,” arXiv preprint, 2023.